

Time-Resolved Optically Detected Magnetic Resonance of Nitrogen-Vacancy Centers in Diamond

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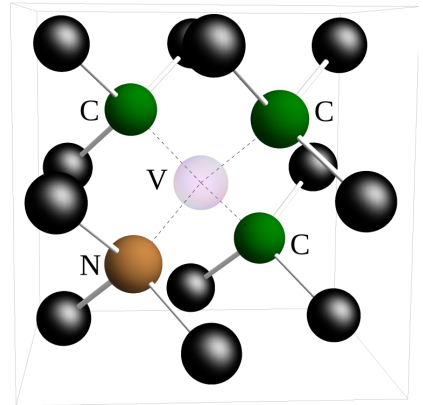
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Motivation: NV Centers in Diamond

The Nitrogen-Vacancy (NV) Center

- Point defect: substitutional nitrogen adjacent to a vacancy.
- Possesses C_{3v} symmetry along the $[111]$ crystallographic axis.
- Ground state is a spin triplet ($S = 1$) formed by localized unpaired electrons.
- Optically active: allows initialization and readout of single spins at room temperature.

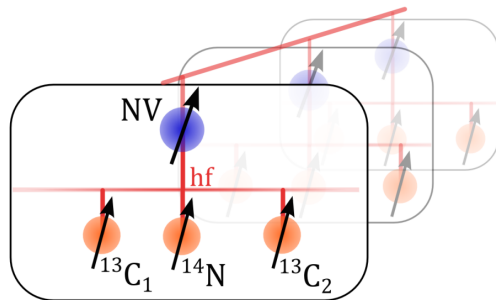


Reference: Doherty et al., Phys. Rep. 528 (2013)

Motivation: The Hybrid Spin Register

Hybrid Spin

- The NV electron spin enables fast, ns-scale microwave control.
- However, it couples to magnetic noise, limiting coherence (T_2).
- **Solution:** Use the electron for readout and control. Use nuclear spins for long-lived coherence (^{14}N , ^{13}C).
- Nuclear spins serve as long-lived quantum memory (seconds to minutes).



Reference: Waldherr et al., Nature 506 (2014)

Theoretical Background: Spin Hamiltonian

The effective $S = 1$ ground-state Hamiltonian:

$$\frac{H}{h} = DS_z^2 + E(S_x^2 - S_y^2) + \gamma_e \mathbf{B} \cdot \mathbf{S} \quad (1)$$

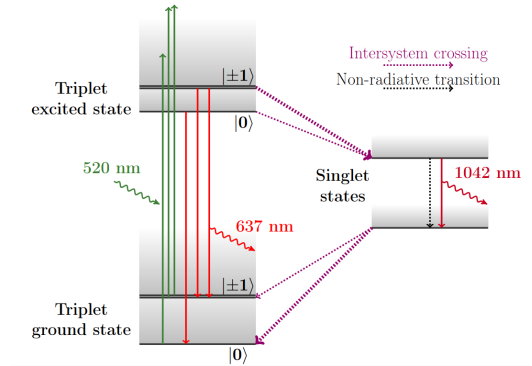
Key Parameters:

- $D \approx 2.87$ GHz: Zero-field splitting from magnetic dipole-dipole interactions.
- E : Strain splitting parameter. Non-axial local lattice strain breaks the C_{3v} symmetry, lifting the degeneracy of the $m_s = \pm 1$ states.
- $\gamma_e \approx 2.8$ MHz/G: Electron gyromagnetic ratio (Zeeman effect).
- $\mathbf{B}$: Applied external magnetic field.

Theoretical Background: Optical Cycle and ODMR

Spin-Dependent Intersystem Crossing (ISC)

- **Excitation:** 532 nm laser pumps $|g, m_s\rangle$ to $|e, m_s\rangle$.
- **Radiative decay:** $|e, 0\rangle \rightarrow |g, 0\rangle$ primarily emits red fluorescence.
- **Non-radiative ISC:** $|e, \pm 1\rangle$ states have a high probability of crossing to $|s\rangle$, which then decays preferentially into $|g, 0\rangle$.
- **Result:** Initialization to $|g, 0\rangle$.

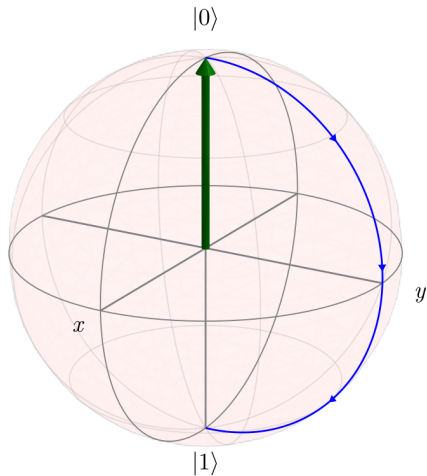


Coherent Spin Manipulation

- **Resonance:** MW field drives transitions from initialized $|g, 0\rangle$ to $|g, -1\rangle$ (or $+1$) only when tuned exactly to the energy gap.

$$\Delta E = \hbar\omega_{\text{MW}} = \hbar(\omega_D \pm \gamma_e B_z)$$

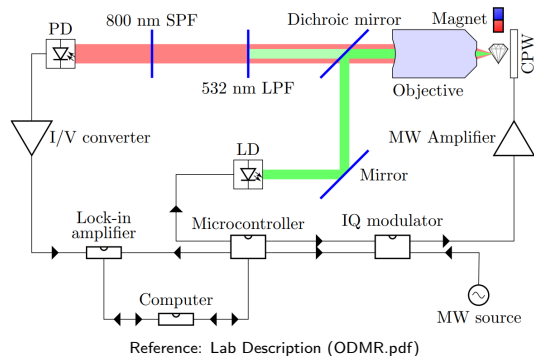
- **Rabi Oscillations:** In the dark, the resonant MW field coherently rotates the state vector between the poles with ω_{Rabi} .



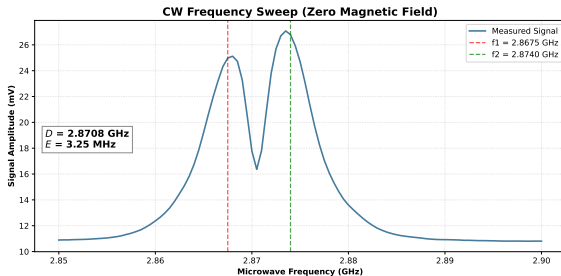
Experimental Methods

Setup and Measurement

- Confocal microscopy used for optical excitation and fluorescence collection.
- Microwave (MW) antenna delivers H -field to drive $\Delta m_s = \pm 1$ transitions.
- **Lock-in Detection:** MW is amplitude-modulated. Lock-in amplifier extracts fluorescence changes precisely at the modulation frequency, rejecting background light drift.



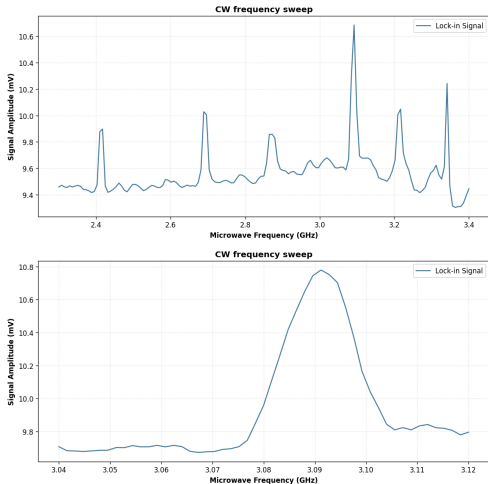
Continuous Wave ODMR: Zero Magnetic Field



Zero-Field Splitting Extraction

- Two resonance peaks observed due to E.
- Measured parameters:
 - $D = 2.87$ GHz (consistent with literature)
 - $E = 3.25$ MHz (consistent: in the MHz range)
- Resonant MW ($D \pm E$) drives population: $|g, 0\rangle \rightarrow |g, \pm 1\rangle$.
- Shifts to dark states $\rightarrow$ steady-state fluorescence drops.

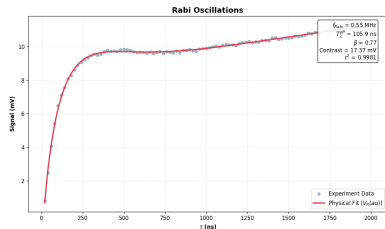
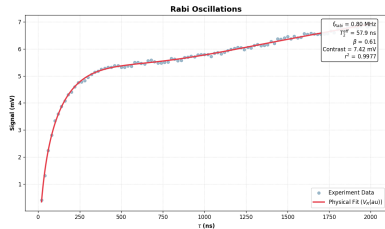
Continuous Wave ODMR: Applied Magnetic Field



Zeeman Splitting

- External $\mathbf{B}$ -field increases degeneracy with $\gamma_e \mathbf{B} \cdot \mathbf{S}$.
- 4 NV orientations yield 8 distinct transitions. We clearly observe 6.
- Focused sweep to isolate a 1 resonance frequency at 3.0912 GHz.
- This frequency targets an isolated two-level subspace ($m_s = 0 \leftrightarrow +1$ or $m_s = 0 \leftrightarrow -1$) for time-resolved control. Becomes a two-state system.

Time-Resolved ODMR: Rabi Oscillations (Our Setup)



$$V_R(\tau) = Ae^{-(\tau/T_2^{\text{eff}})^{\beta}} \sin(\omega_{\text{Rabi}}\tau + \phi) + B\tau + C$$

- ω_{Rabi} : Rabi frequency.
- T_2^{eff} : Effective dephasing time.
- β : inhomogeneity.
- B : temperature effects due to the MW power.
- Could not measure ω_{Rabi}
- Optimized magnet and sample position for higher signal amplitude
- Extracted $T_2^{\text{eff}} \approx 100 \text{ ns}$ in the bottom graph.

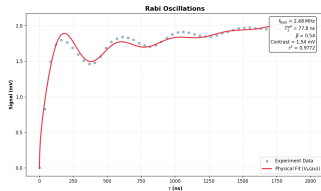
Time-Resolved ODMR: Rabi Oscillations

Extracted Rabi Frequencies

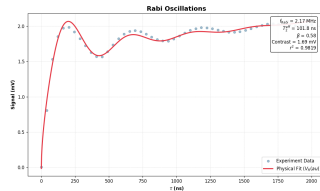
- **0 dB:** $f_{\text{Rabi}} = 2.5$ MHz
- **6 dB:** $f_{\text{Rabi}} = 2.1$ MHz
- **10 dB:** $f_{\text{Rabi}} = 1.5$ MHz

Power Dependence

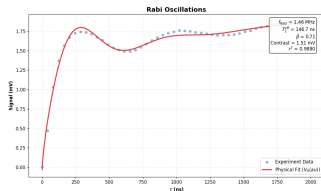
- Using an attenuator, the power dependence could be analyzed.



0 dB attenuation



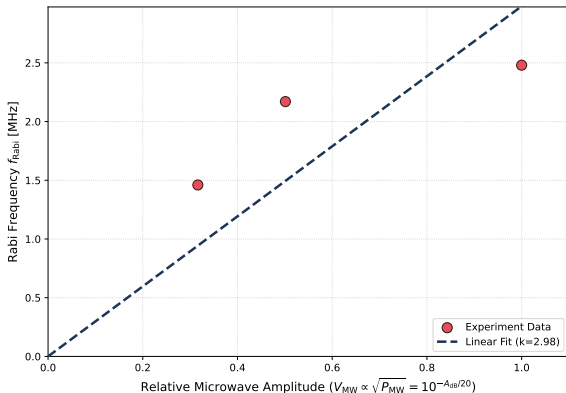
6 dB attenuation



10 dB attenuation

Rabi Frequency Power Dependence

Rabi Frequency Power Dependence



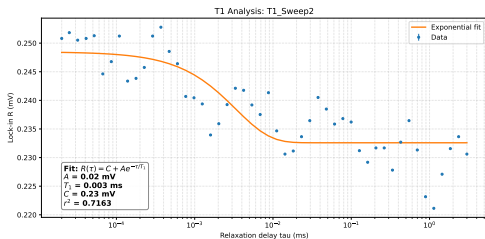
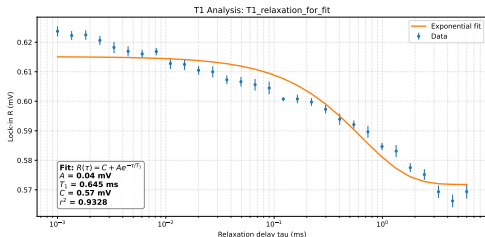
Driving Amplitude Scaling

- Theory: $\omega_{\text{Rabi}} \propto B_{\text{MW}} \propto \sqrt{P}$.
- Dephasing:
 - Higher MW power decreases T_2^{eff} .
 - Caused by spatially inhomogeneous MW field.
 - Slightly different Rabi frequencies $\rightarrow$ destructive interference.

Longitudinal Relaxation vs. ISC

$$V_{T_1}(\tau) = A \exp(-\tau/T_1) + C$$

- Measure population recovery during dark time τ .
- We expected ms range T_1
- **Analysis of measured 645 ns decay:**
 - True T_1 (spin-lattice relaxation) requires milliseconds at room temperature.
 - The measured ≈ 600 ns timescale represents the *intersystem crossing (singlet state) lifetime*.
 - Captures the optical pumping transient, not dark thermalization.



Summary and Conclusions

- **System Characterization:** Successfully mapped the zero-field splitting ($D = 2.87$ GHz) and identified local strain limits ($E \approx 3.25$ MHz).
- **Zeeman Control:** Isolated a specific $m_s = 0 \leftrightarrow -1$ transition at 3.0912 GHz for time-resolved coherent manipulation.
- **Coherent Dynamics:** Observed Rabi dephasing ($T_2^{\text{eff}} \approx 100$ ns) and validated the $\omega_{\text{Rabi}} \propto \sqrt{P}$ scaling law using complementary data.
- **Relaxation Timescales:** Differentiated between the fast non-radiative singlet decay transient (~ 600 ns) and long-term spin-lattice T_1 relaxation dynamics.

Thanks for the Attention

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